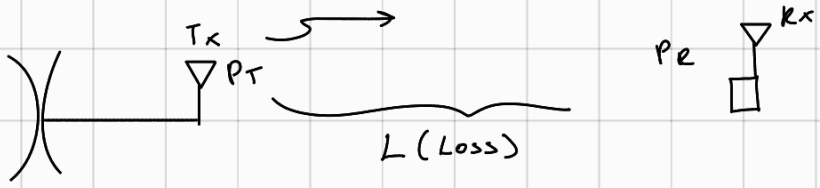
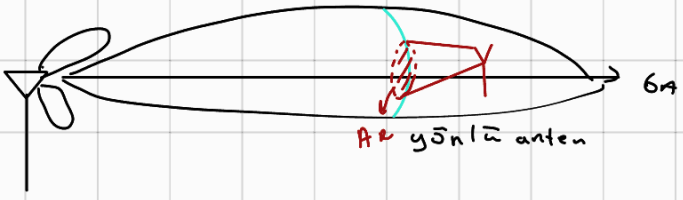
RF link

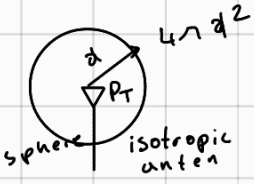


1) Antenna Gain

Antenna efficiency $\quad \eta = \frac{P_T}{P_{in}} < 1$

$G_A =$ Maximum power density of the actual antenna / power density of isotropic antenna





$4\pi d^2$ = sphere area

eğer P_T gücünü yayıyorsa

$$P_D \text{ (power density)} = \frac{P_T}{4\pi d^2}$$

Antenler her yöne eşit dağılım yapmaz. Bu nedenle her yöne eşit dağılım yaptığını varsaydığınız isotropic antenle mukayese ederiz

$G_A \rightarrow$ if antenna has gain $\quad P_D = \frac{P_T \cdot G_A}{4\pi d^2}$

if antenna has Gain and efficiency $\rightarrow P_D = \frac{P_T G_A \cdot \eta}{4\pi d^2}$ $[W/m^2]$

→ Güç yoğunluğu mesafenin karesiyle ters orantılı olarak değişiyor.

Alıcıda: for Rx $\longrightarrow A_R = \frac{G_R \cdot \lambda^2}{4\pi}$ $[m^2]$ (λ: wavelength) → Effective Arpeture Size (Etkin açıklık)

$$P_R = P_D A_r$$

$$P_R = \frac{P_T G_A \cdot \eta_A}{4\pi d^2} \cdot \frac{G_R \cdot \lambda^2 (\eta_r)}{4\pi}$$

$$P_R = P_T \cdot G_A \cdot G_R \cdot \eta_A \cdot \eta_R \left(\frac{\lambda}{4\pi d}\right)^2$$

(G_A: Tx antenna, G_R: Rx antenna)

$$P_R|_{dBm} = P_{T\,dBm} + G_{A\,dB} + G_{R\,dB} + \eta_A|_{dB} + \eta_D|_{dB} + \underbrace{20\log\left(\frac{\lambda}{4\pi d}\right)}_{L_{path,loss}(LOS)}$$

$L_{link\,dB}$ (LOS) → Alıcıyla vericinin birbirini görd=ğ= durumu için

Line of side

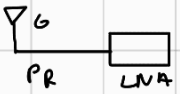
Ex: point-point (LOS)

$G_R = 60$ dB

$f = 60$ GHz

$P_{R\,density} = 1\,pW/cm^2$

what is the power at output of antenna connected to RF electronics

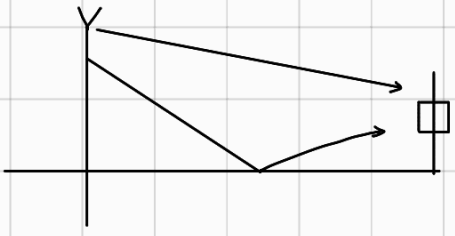


$\lambda = \frac{c}{f}$

$$A_R = \frac{G_R \cdot \lambda^2}{4\pi} = \frac{10^6\left(\frac{3\cdot10^8}{60\cdot10^9}\right)^2}{4\pi} = 1{,}99\,m^2 \approx 2\,m^2$$

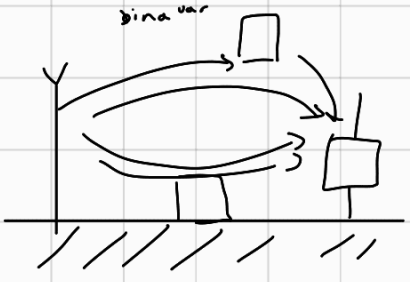
$$P_R = P_D \cdot A_R = \frac{10^{-12}}{cm^2} \cdot 2\cdot10^4\,cm^2 = 20\,nW$$

LOS olmazsa



direct signal is dominant

Ricean fading $\left[\frac{1}{d^3}\right]$



→ reflected waves / Rayleigh fading $\left[\frac{1}{d^4}\right]$

$$L_{path,loss}(LOS) = \left(\frac{4\pi d}{\lambda}\right)^2$$

$$L_{path,loss}(NLOS) = \left(\frac{4\pi d}{\lambda}\right)^2 (d)^{n-2}$$

(NLOS): non line of side

n: number of paths

$$L_{path,NLOS}|_{dB} = 20\log\left(\frac{4\pi d}{\lambda}\right) + 10(n-2)\log(d)$$

$$= 10n\log(d) + 20\log\left(\frac{4\pi}{\lambda}\right)$$

Ex: $f = 5.6\,GHz$

$G_A = 35\,dB$

$G_R = 6\,dB$

$d = 200\,m$

$\frac{1}{d^3} \rightarrow n = 3$

$$L_{Link} = -G_A - G_R + 10\,n\log(d) + 20\log\left(\frac{4\pi}{\lambda}\right)$$

$$= -35 - 6 + 10.3\log(100) + 20\log\left(\frac{4\pi}{\frac{3.10^8}{5,6.10}}\right)$$

$$= 75,4\,dBm$$

Link Margin

$$P_R = P_T - L_{Link}$$

$$LM = P_R - S$$ (S: alıcı hassasiyeti)

Link Margin

Bu linkte oluşabilecek bozulmalara 3dBm bir pay bulunmaktadır. Eğer 5dBm'lik bir bozulma olursa alıcı artık çalışmıyordur (3 - 5 = -2dBm)

$S = -100\,dBm$

$P = -97\,dBm$

$LM = -97 - (-100)$

$LM = 3dB$

Link Budget Table

Tx power	P_T	$\longrightarrow$ 10dBm
Gains	G_T	$\longrightarrow$ 10dB
	G_R	$\longrightarrow$ 5dB
Loss	η_T	$\longrightarrow$
	η_R	$\longrightarrow$
Link loss	L_{Link}	$\longrightarrow$ ...
Received Power	P_R	$\longrightarrow$

$LM = P_R - S$